```
    print("Bulb Remains OFF")

    GPIO.output(RELAY_PIN, GPIO.LOW)

    sleep(2)
```

**7. Result:**

## PRACTICAL – 13

**1. Aim**

Implement basic Publisher – Subscriber using MQTT. Publish messages using mqtt. Subscribe and control LED.

**2. Prior Concepts**

i) Basic Raspberry Pi architecture and GPIO programming.

ii) Client–Server communication.

iii) Command-line operations in Linux.

**3. New Concepts**

i) MQTT publish-subscribe communication model.

ii) Working of MQTT broker, publisher and subscriber.

iii) Using Paho MQTT library in Python.

iv) Event-driven programming using MQTT callbacks.

v) Building cloud-ready IoT applications.

**4. Hardware Required**

i) LED

ii) 220 Ω Resistor

iii) Jumper Wires

**5. Procedure**

**Pin Connection**

**LED Connection**

| **LED Pin** | **Raspberry Pi 4** |
|---|---|
| Anode (+) | GPIO18 (Pin 12) through 220Ω resistor |
| Cathode (-) | GND (Pin 6) |

# Input IMAGE

i) Install Python3 on Raspi.

Write the below code in command prompt:

**sudo apt install idle3**

Install MQTT Broker:

**sudo apt update**
**sudo apt install mosquitto mosquitto-clients -y**

Enable broker:

**sudo systemctl enable mosquitto**
**sudo systemctl start mosquitto**

Install Paho MQTT:

**pip3 install paho-mqtt**

**6. Implementation with Python Code**

**Subscriber Code (LED Control):**

```
import paho.mqtt.client as mqtt
import RPi.GPIO as GPIO

LED = 18

GPIO.setmode(GPIO.BCM)
GPIO.setup(LED, GPIO.OUT)

BROKER = "localhost"
TOPIC = "iot/led"

def on_connect(client, userdata, flags, rc):
    print("Connected")
    client.subscribe(TOPIC)

def on_message(client, userdata, msg):

    message = msg.payload.decode()

    print("Received:", message)

    if message.upper() == "ON":
        GPIO.output(LED, GPIO.HIGH)
        print("LED ON")

    elif message.upper() == "OFF":
        GPIO.output(LED, GPIO.LOW)
        print("LED OFF")
```

```
client = mqtt.Client()

client.on_connect = on_connect
client.on_message = on_message

client.connect(BROKER, 1883, 60)

client.loop_forever()
```

**Publisher Code:**

```
import paho.mqtt.client as mqtt

broker = "localhost"
topic = "iot/led"

client = mqtt.Client()

client.connect(broker, 1883, 60)

while True:

    msg = input("Enter ON/OFF : ")

    client.publish(topic, msg)

    print("Published :", msg)
```

**7. Execution Step**

Terminal 1: Run Subscriber

**python3 subscriber.py**

Terminal 2: Run Publisher

**python3 publisher.py**

**8. Result**

**Subscriber Terminal:**

Connected
Received: ON
LED ON
Received: OFF
LED OFF

**Publisher Terminal:**

text
Enter ON/OFF : ON
Published : ON
Enter ON/OFF : OFF
Published : OFF

## PRACTICAL – 14

**1. Aim**

Design a web dashboard using Flask. Control relay module via web.

**2. Prior Concepts**

i) Concept of Python Programming.

ii) Working of relay modules.